Acute Otitis Media

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Introduction

More than 80% children experience at least one episode of acute otitis media (AOM) by age 3. Viral upper respiratory tract infection is the most common predisposing factor. *Streptococcus pneumoniae*, nontypeable *Haemophilus influenzae,* and *Moraxella catarrhalis* are the most common bacterial etiological agents. Viral agents include respiratory syncytial virus, picornavirus, coronavirus, influenza viruses, and adenovirus. Symptoms may include ear pain (rubbing, tugging, or holding the ear may be a sign of pain), fever, irritability, otorrhea, anorexia, and sometimes vomiting or lethargy.

Diagnosis

Acute otitis media is diagnosed in symptomatic children who present with:

- ☑ Moderate-to-severe bulging of the tympanic membrane or new-onset otorrhea not caused by acute otitis externa
- ☑ Mild bulging and either recent-onset ear pain (<48 hours) or intense erythema of the tympanic membrane.
  Diagnosis should not be made in children without middle ear effusion (MEE).

Otoscopy should be used for the assessment of the tympanic membrane. Pneumatic otoscopy is up to 94% sensitive and 90% specific for identifying MEE.

Treatment

Symptomatic Therapy

Symptomatic therapy with oral ibuprofen or paracetamol is used for ear pain. Decongestants and antihistamines have unproven role in symptom relief. Topical anesthetic eardrops and naturopathic eardrops have been found to decrease pain in some small studies, but overall evidence is insufficient to recommend routine use. They should be avoided if there is any concern for perforated tympanic membrane.

Indications of Antibiotics

The decision of starting with an antibiotic depends on the laterality, presence of risk factors, age of the child, and severity of illness.

- ☑ For children <6 months, antibiotic should be started immediately (even with presumed episode of AOM).
- ☑ For children 6 months to 2 years, with confirmed unilateral or bilateral AOM of any severity should be started on antibiotics.
- ☑ For children >2 years, with confirmed AOM, severe disease (i.e., temperature >39°C (102.2°F) in the past 48 hours, significant otalgia, or toxic appearance) should be immediately started with an antibiotic.

Consider deferring antibiotic therapy or "watchful waiting" for 2–3 days with close follow-up in children >2 years with nonsevere otitis media (OM) or questionable diagnosis and in children 6 months to 2 years with questionable diagnosis and nonsevere disease.

Initial Anti-microbial Therapy

Choice of drug depends on its clinical and microbiological efficacy, acceptability of oral preparation, side effects, convenience of dosing schedule, and cost **(Flowchart 1 and Table 1)**.

Flowchart 1: Algorithm for acute otitis media (AOM) management requiring antibiotics.

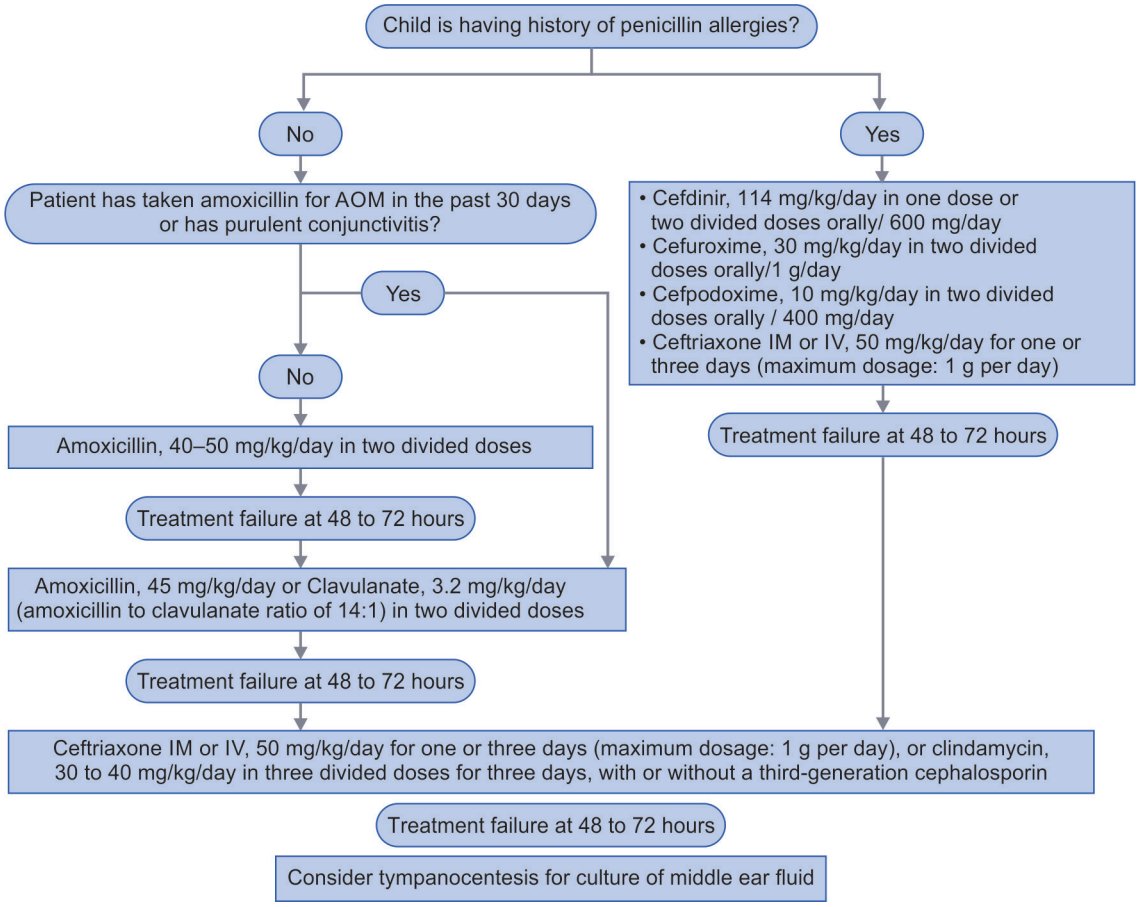


TABLE 1: Alternatives for cases of severe reaction to beta-lactams including cephalosporins.

Azithromycin	10 mg/kg once on day 1, then 5 mg/kg once a day from day 2 to 5, orally	500 mg on day 1, 250 mg on days 2–5
Clarithromycin	15 mg/kg/day in two doses	1 g/day
Clindamycin	20–30 mg/kg/day in three doses	1.8 g/day

Note: For cases of persistent treatment failure, a referral to pediatric otolaryngologist and/or pediatric infectious disease expert is recommended.

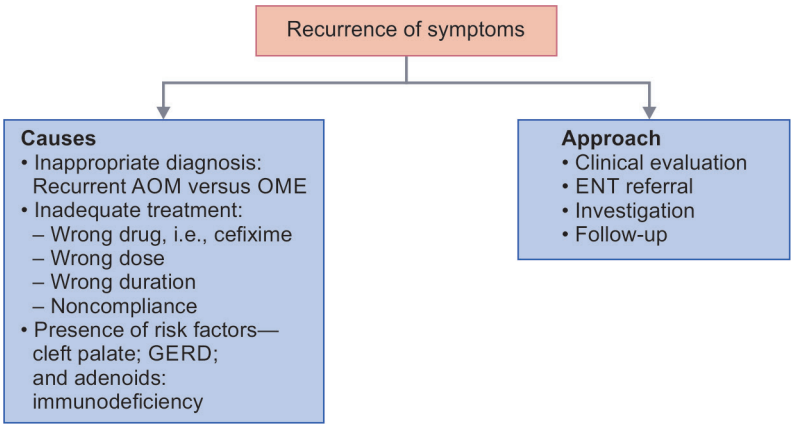
Treatment

Duration of Treatment

- ☑ For amoxicillin, amoxicillin-clavulanate, oral cephalosporins, clarithromycin, and clindamycin:
  - Children <2 years—10 days
  - Children >2 years—5–7 days
- ☑ For ceftriaxone—one to three doses, depending on persistence of symptoms
- ☑ For azithromycin—5 days

Recurrent Acute Otitis Media

It is defined by development of signs and symptoms of AOM within 30 days of completion of successful treatment.



(AOM: acute otitis media; ENT: ear nose throat; GERD: gastroesophageal reflux disease; OME: otitis media with effusion)

Indications of Tympanostomy Tube Insertion

For children with more than three episodes of AOM within 6 months or more than four episodes within 12 months with one episode in preceding 6 months.

Further Reading

- ☑ American Academy of Pediatrics Subcommittee on Management of Acute Otitis Media. Diagnosis and management of acute otitis media. Pediatrics. 2004;113:1451-65.
- ☑ Gupta P, Menon PSN, Ramji S, Lodha R. PG Textbook of Pediatrics, 2nd edition. New Delhi: Jaypee Brothers Medical Publishers; 2018.
- ☑ Kliegman RM, St Geme J. Nelson Textbook of Pediatrics, 21st edition. Philadelphia, PA: Elsevier; 2019.
- ☑ Lieberthal AS, Carroll AE, Chonmaitree T, Ganiats TG, Hoberman A, Jacksonet MA, et al. The diagnosis and management of acute otitis media. Pediatrics. 2013;131:e964-99.

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Acute Rheumatic Fever

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